Concentric Cubes Are Plumbing, Not Security

Design and Cryptanalysis of `cube3d-sponge`, a Three-Dimensional Cellular-Automaton Sponge Permutation

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Abstract

A recurring intuition in the cellular-automaton (CA) cryptography literature holds that adding spatial dimensions, neighbourhood radius, or nested structure to a CA makes the resulting cipher "more complex and hence more secure." We take that intuition seriously enough to test it. Starting from a proposal to fold plaintext into a three-dimensional grid of concentric cubic shells and evolve a CA radiating outward from the centre, we show that the geometric intuition is *orthogonal* to security: apparent complexity contributes nothing that is not already captured by nonlinearity, invertibility, symmetry-breaking, and diffusion uniformity. We then show that the concentric-shell geometry *does* have one legitimate cryptographic role — as a sponge rate/capacity boundary — and build `cube3d-sponge`, an $8 \times 8 \times 8$ byte-state (4096-bit) permutation combining a checkerboard generalised Feistel network over a 3-torus, a χ-family nonlinear map, a per-cell byte rotation, and a coordinate shear, wrapped in a sponge whose rate is the outer cubic shell and whose capacity is the inner shells.

Our contribution is primarily *negative and methodological.* We report a complete cryptanalytic ladder against the construction, obtained from four independent harnesses and cross-validated on two operating systems: key recovery by cube attack succeeds only to round 2 of 12; deterministic zero-sum distinguishers derived from provable degree upper bounds reach round 4 with practical cubes (2^{16}) and round 5 with a 2^{64} cube; statistical differential bias is detectable to round 6. We document two design bugs the harnesses caught that would otherwise have produced confident nonsense — a checkerboard 2-colouring that is inconsistent on an odd-length torus, and bitwise operations that left the eight bit-planes of the state *provably non-interacting*, capping avalanche at exactly $512/2/4096 = 6.25\%$ — and two harness bugs that produced impossible (non-monotonic) measurements before being fixed. We argue that the discipline of building the falsifier before trusting the design is the transferable result here, and we explicitly decline to claim that `cube3d-sponge` is secure. It is merely *not obviously broken*, which is a far weaker statement, and we recommend Ascon or ChaCha20-Poly1305 for any real use.

Contents

1 Introduction

1.1 The question

Cellular automata are seductive to cryptographers. They are local, parallel, cheap in hardware, and they generate visually chaotic output from trivially simple rules. It is tempting — and the literature shows it has been tempting for forty years — to read that visual chaos as cryptographic strength.

This work began with a concrete proposal in that tradition: given prior work analysing data plotted as two-dimensional grids under CA evolution, would folding data into a *three-dimensional* grid — a central cube surrounded by concentric shells of smaller cubes, with the CA radiating outward in all directions — make the technique "way more complex and hence more secure"?

The question deserves a precise answer, because the reasoning inside it is the single most common failure mode in the CA- and chaos-cryptography literature. The answer has three parts:

1. **No, not as stated.** Complexity is not security. Security is the analysed work factor for an adversary who knows the entire construction and lacks only the key (Kerckhoffs [1]). Adding dimensions, shells, or radius adds state and cost; it does not add any of the properties that actually resist cryptanalysis. A larger linear system is still solved by Gaussian elimination.

2. **But the shape is right.** A 3D local nonlinear permutation iterated with round constants is, almost exactly, Keccak [21]. Keccak's state *is* a $5 \times 5 \times 64$ cube of bits. The difference between Keccak and a naive radiating CA is a small, enumerable set of properties — not the geometry.

3. **And one piece of the geometry earns its keep.** Concentric shells are a natural *rate/capacity boundary*: outer shell public, inner shells never directly output. That is the sponge construction [20], and it is a legitimate, analysed structural role for "layers."

1.2 What we did

We built the construction that (3) suggests, and then we attacked it. The point was never to produce a competitive cipher — we say plainly in Section 9 that nobody should use this — but to find out what a disciplined cryptanalytic process reports about a design whose provenance is "geometric intuition, corrected."

Specifically, we contribute:

- **A design** (Section 3) that isolates the geometric idea from the security-relevant machinery: concentric shells serve *only* as the rate/capacity split, while diffusion uniformity is handled independently by toroidal wrapping. Conflating those two roles — which the original "radiate from the centre" framing does — makes shell boundaries simultaneously diffusion weak points and security boundaries.
- **A cryptanalytic ladder** (Section 7) from four independent harnesses: avalanche/diffusion, statistical differential and linear distinguishers, a cube attack with real key recovery, and provable algebraic degree bounds yielding deterministic zero-sums.
- **A record of the bugs** (Sections 4 and 6), both in the design and in the harnesses. We consider this the most transferable part of the paper. Every one of the four was caught by a measurement that produced a *physically impossible* number, not by inspection.
- **An explicit refusal to overclaim.** Upper bounds on degree prove that distinguishers exist; they never prove that they do not. Negative attack results mean "this attack, at this budget, found nothing."

1.3 A note on a corrected novelty claim

Early in this work we believed the concentric-cube geometry to be structurally novel. A literature search conducted for this paper shows that belief was too strong, and we correct it here rather than quietly dropping it. Martín del Rey [17] encrypts 3D solid objects using a three-dimensional chaotic Cat map for confusion followed by a *second-order memory reversible 3D cellular automaton* for diffusion, iterated over several rounds — which is, almost point for point, the architecture we independently proposed as the "principled path." 3D CA image and object encryption is an established sub-literature [17, 11, 18, 19]. What remains arguably new in our construction is narrower: the checkerboard Feistel over a 3-torus as an invertibility mechanism, and the use of concentric shells specifically as a sponge rate/capacity boundary rather than as a diffusion structure. We make no strong novelty claim beyond that, and we note that novelty of geometry would not, in any case, imply novelty of security argument.

2 Related work

2.1 Cellular-automaton cryptography and its cryptanalytic record

Wolfram proposed elementary CA rule 30 as the basis of a Vernam-like stream cipher at CRYPTO 1985 [3], motivated by its Class III chaotic dynamics: the key is the initial configuration of a ring of cells, and the sequence of states of a distinguished cell is the keystream. Meier and Staffelbach broke it at EUROCRYPT 1991 [4] with a correlation attack that reconstructs the seed despite the sequence passing classical statistical randomness tests. This is the canonical lesson of the field, and it is worth stating in its strongest form: *passing statistical tests is not evidence of cryptographic strength.* Rule 30 passed; rule 30 fell.

Subsequent work has attempted to repair the approach — by enlarging the neighbourhood radius [8], by using maximum-period nonlinear CA [7], by hybridising linear and nonlinear rules — with mixed and often short-lived success. Mariot and Leporati's recent survey [6] of a decade of CA-based cryptography is candid about the field's methodological problems, noting in particular that most work relies on ad hoc arguments specific to individual rules that generalise poorly.

It is worth recording the one unambiguous success story: the χ mapping, a degree-2 shift-invariant nonlinear CA-like transformation introduced by Daemen et al. [5], is the nonlinear core of Keccak [21] and, in affine equivalent form, of Ascon [23]. CA ideas did produce a standards-grade primitive — but as *one layer* inside a carefully analysed round function, never as the whole cipher.

2.2 The chaos/CA image-encryption literature

The proximate context for the original question is the large literature on chaos- and CA-based image encryption. Its architectural template is due to Fridrich [9]: a permutation (confusion) stage scrambles pixel *positions* using a 2D or 3D chaotic map, followed by a diffusion stage that modifies pixel *values*, with the pair iterated for several rounds. This directly instantiates Shannon's confusion/diffusion principle [2].

The cryptanalytic record of this literature is poor, and it is important to state so plainly because sheer publication volume can be mistaken for validation. Fridrich's own founding scheme was cryptanalysed by Solak et al. [10], with an attack that extends to structurally similar designs including the widely-cited 3D chaotic Cat map scheme of Chen et al. [11]. Li and collaborators have broken a long series of such schemes [13, 14, 16, 15], frequently with a handful of chosen plaintexts. Most damningly, Li, Chen and Lo [12] give a *generic* result: all permutation-only multimedia ciphers are practically insecure against known/chosen-plaintext attack, requiring only $O(\log_L(MN))$ plaintexts to recover at least half of a plaintext of size $M \times N$ with L value levels.

Two conclusions follow, both directly relevant to the question this paper answers. First, "cascading more chaotic maps / more CA layers" does not compose weak primitives into a strong one. Second — and this is the specific trap in the original proposal — a *fixed, public* folding of data into a grid is a known permutation, and a known permutation contributes exactly zero entropy.

2.3 Three-dimensional CA cryptography

Prior 3D CA cryptography exists and is not marginal. Beyond Rey [17] discussed in Section 1.3, there is work rendering images into 3D lattices with chaotic Cat map confusion and 3D CA diffusion [18], 3D image encryption via integral imaging and CA transforms, and a 2026 review covering 1D, 2D and 3D CA image encryption comprehensively [19]. The recurring pattern is that the third dimension is used as *extra scrambling volume.* We are not aware of prior work using concentric shells of a 3D CA state as a sponge rate/capacity boundary, which is the one structural element we would defend as new — while noting again that structural novelty is not a security argument.

2.4 Sponge constructions

The sponge construction [20] maintains a state split into a *rate* r (absorbed into and squeezed from) and a *capacity* c (never touched directly by the outside world), with $b = r + c$. Security against generic attacks is governed by c, not b. Keccak [21], standardised as SHA-3 [22], instantiates this with a $b = 1600$-bit permutation whose state is a $5 \times 5 \times 64$ array of bits, and whose round is $\iota \circ \chi \circ \pi \circ \rho \circ \theta$. Ascon [23], selected by NIST for lightweight cryptography and specified in SP 800-232 [25], uses a 320-bit sponge/duplex with a 5-bit S-box that is an affine equivalent of Keccak's χ, applying 12 rounds for initialisation/finalisation and 8 for the data path.

The relevance to the original question is direct. Keccak is a three-dimensional, local, nonlinear, iterated permutation — the intuition "operate a 3D local rule repeatedly on a cube of data" is not wrong, it is *already standardised*. What separates Keccak from a naive radiating CA is a short list: a nonlinear step (χ), a symmetry-breaking round constant (ι), a bit-transposition step (ρ, π) and a linear diffusion step (θ) — plus, decisively, a decade of adversarial analysis.

2.5 Cryptanalytic tools

Cube attacks. Dinur and Shamir [26] observe that any output bit of a keyed primitive is a polynomial over $\mathbb{F}_2$ in key and public (IV/nonce) variables. Summing the output over all 2^d assignments of a chosen d-subset I of public variables (the *cube*) annihilates every monomial not divisible by the cube monomial t_I, leaving the *superpoly* $p_{S(I)}$. If the total degree is at most $d+1$, the superpoly is linear in the key bits, yielding a free linear equation. Enough independent linear superpolys recover the key by Gaussian elimination. Linearity is tested probabilistically by the Blum–Luby–Rubinfeld test [28]: a linear p satisfies $p(x) \oplus p(y) \oplus p(0) = p(x \oplus y)$. Cube testers [27] extend the idea to distinguishers.

Division property. Todo [29] introduced the division property as a generalised integral property; Todo and Morii [30] refined it to the bit-based division property (BDP). Xiang et al. [31] made it practical at realistic block sizes by modelling division-trail propagation as Mixed Integer Linear Programming, solved with off-the-shelf solvers. Division property gives the current state of the art in *provable* degree upper bounds and zero-sum distinguishers, and Hu et al.'s monomial-prediction formulation [34] makes degree evaluation exact.

Numeric mapping. Liu [33] introduced numeric mapping, a cheap degree-propagation technique for NFSR-based cryptosystems. It propagates per-bit degree upper bounds through $\deg(a \oplus b) \leq \max(\deg a, \deg b)$ and $\deg(a \wedge b) \leq \deg a + \deg b$. It is sound but loose. Zhang et al. [35] compare Boura–Canteaut's formula [36], Carlet's formula, Liu's numeric mapping and Todo's division property, proving division property is never worse than the first two. Hu et al. [34] quantify the gap dramatically: the exact degree of 834-round Trivium is 78, whereas Liu's numeric mapping bound already reaches 79 at 793 rounds. We use numeric mapping in this work, and Section 9 is explicit that this is a tractability compromise, not a choice of the best tool.

Degree bounds. Canteaut and Videau [37] bound the degree of a composition; Boura, Canteaut and De Cannière [36] give bounds specialised to Keccak-like permutations that are tighter than the naive $(\deg F)^R$ near saturation.

3 Design of `cube3d-sponge`

3.1 State geometry

The state is an $8 \times 8 \times 8$ array of bytes: 512 cells, 4096 bits. Cell (x, y, z) has index $i = x+8y+64z$.

Why $N = 8$ and not 7. Our first implementation used $N = 7$ to obtain a single centre cell, matching the "central cube" of the original proposal literally. This is *mathematically broken*, and the unit test caught it immediately. The round function (below) requires a checkerboard 2-colouring by the parity of $x+y+z$ such that every face-neighbour has the opposite colour. On a torus with wrap-around, this holds only if the side length is even: an odd-length cycle is not bipartite. With $N = 7$, cells at $x = 0$ and $x = 6$ are wrap-around neighbours and share a colour, silently invalidating the Feistel invertibility argument. With $N = 8$ the colouring is consistent, and as a bonus the centre is a literal $2 \times 2 \times 2$ cube of 8 cells — closer to the original geometric picture than a single point.

Shells. Since N is even there is no single centre coordinate, only a centre *pair* $\{3, 4\}$ per axis. Define the per-axis shell $s(v) = \min(|v-3|, |v-4|)$ and the cell shell $\mathrm{rad}(i) = \max(s(x), s(y), s(z)) \in \{0, 1, 2, 3\}$. Shell 0 is the central $2 \times 2 \times 2$ cube; shell 3 is the outer face shell.

Rate and capacity. The outer shell (rad $= 3$, 296 cells) is the sponge **rate**: the only cells `absorb` and `squeeze` ever touch. Shells 0–2 (216 cells, 1728 bits) are the **capacity**: never read or written directly. This is the legitimate role of "concentric layers" promised in Section 1.

Colouring. $\mathrm{col}(i) = (x + y + z) \bmod 2$. On the even torus, every face-neighbour of a cell has the opposite colour.

3.2 Round function

One round is $\pi \circ \rho \circ$ Feistel. Each step is individually invertible, so the composition is a permutation by construction.

χ_3: the nonlinear step. For cell i with toroidal face-neighbours $(n_0, \ldots, n_5)$ and round r, let $j_k = n_{(k+2(r \bmod 3)) \bmod 6}$. Then

$$\chi_3(S, i, r) \;=\; (\neg S_{j_0} \wedge S_{j_1}) \oplus (\neg S_{j_2} \wedge S_{j_3}) \oplus (\neg S_{j_4} \wedge S_{j_5}),$$

computed bytewise. This is in the χ family of Daemen et al. [5]: it has algebraic degree 2, and it is the only source of nonlinearity in the design. Neighbours are *toroidal*, which gives every cell an identical neighbourhood shape; without wrapping, shell-boundary cells would diffuse differently from interior cells, which is precisely the anisotropy the original "radiate outward" framing would have introduced.

Feistel: the invertibility mechanism. Because every face-neighbour of an A-cell is a B-cell, χ_3 evaluated at an A-cell reads *only* B-cells. This is exactly the condition a Feistel network needs. With A_0, B_0 the colour classes of the input state:

$$A_1[i] = A_0[i] \oplus \chi_3(B_0, i, r), \qquad \text{for all } A\text{-cells } i, \tag{1}$$

$$A_1' = A_1 \text{ with RC}(r) \text{ XORed into cell } 0, \tag{2}$$

$$B_1[i] = B_0[i] \oplus \chi_3(A_1', i, r), \qquad \text{for all } B\text{-cells } i. \tag{3}$$

Proposition 1 (Invertibility). *The Feistel step is a bijection for* any *function χ_3 whatsoever, provided χ_3 evaluated at a cell of one colour reads only cells of the other colour.*

Proof. Given (A_1', B_1), recover $B_0[i] = B_1[i] \oplus \chi_3(A_1', i, r)$ (which needs only A_1', available); then $A_1 = A_1'$ with RC(r) XORed out; then $A_0[i] = A_1[i] \oplus \chi_3(B_0, i, r)$ (which needs only B_0, now recovered). □

This is why we chose a Feistel over a second-order or Margolus partitioning CA: it decouples the invertibility argument entirely from the choice of nonlinear rule, letting us make χ_3 as nonlinear as we like without ever endangering decryption.

ι-analogue: round constants. RC(r) is a cheap integer-hash of the round index, XORed into cell 0 only (as Keccak's ι touches one lane). It is load-bearing, not decorative: the χ_3 tap rotation has period 3 in r, so *without* the constant, round 0 and round 3 are literally the same function — a textbook slide-attack foothold. Our `slide_test` harness verifies exactly this (Section 7.4). Note also that any constant state $S_i = c$ satisfies $\chi_3(S) = 0$, since $\neg c \wedge c = 0$; without the round constant, all 256 constant states would be fixed points of the Feistel.

ρ: bit-plane coupling. Rotate each cell's byte left by $\rho(i) = (x + 3y + 5z) \bmod 8$. Section 4 explains why this step is the single most important component in the design.

π: transposition. A coordinate shear mod 8:

$$x' = x + y, \quad y' = y + z, \quad z' = z + x' \pmod 8.$$

The system is triangular, so the inverse is immediate: $z = z' - x'$, $y = y' - z$, $x = x' - y$. No lookup table is needed, and bijectivity is verified by unit test rather than asserted. π deliberately moves cells *across* shells; the rate/capacity split is positional, exactly as Keccak's rate is "the first r bits" while π shuffles bits through those positions.

3.3 Mode

`Sponge::absorb` XORs up to 296 bytes into the rate cells and applies permute = 12 rounds; `squeeze` reads rate cells and permutes as needed. `keystream(key, nonce)` absorbs the key, permutes, absorbs the nonce, permutes, and squeezes. *There is no authentication tag.* This is a keystream generator only, and Section 9 says what to do about that (namely: do not use it).

4 Two design bugs, and how they were caught

Both bugs below were caught by measurements returning *impossible* numbers, not by code review. We report them because they are the most transferable content here.

4.1 The odd-torus colouring

Discussed in Section 3: with $N = 7$, the unit test `checkerboard_neighbors_always_opposite_colour` failed instantly. Had we merely eyeballed the design, the invertibility proof would have been silently false for wrap-around neighbours.

4.2 Bit-plane independence: the 6.25% plateau

Our first design had χ_3 and the Feistel but *no* ρ. Avalanche measurement showed the state difference from a single flipped input bit stuck at $\approx 6.3\%$ and *perfectly flat* through 16 rounds — not a slow ramp, a hard ceiling.

The diagnosis is arithmetic. χ_3 uses bitwise `AND`/`NOT`/`XOR` on `u8` cells. Bitwise operations act on all 8 bits in parallel but *independently*: bit k of the output depends only on bit k of the inputs. Without ρ, the "$8 \times 8 \times 8$ cube of bytes" was never a single 4096-bit state at all — it was **eight completely independent, non-interacting 512-bit binary CAs stacked in the same memory**. A flipped bit could never leave its own bit-plane. Hence the saturation ceiling is

$$\frac{512}{2}\Big/4096 = 6.25\%,$$

which is what we measured. Seven-eighths of the state was dead weight *by construction.*

Remark. Our initial hypothesis was wrong, and instructively so. We first attributed the plateau to insufficient long-range transposition. That diagnosis is refuted by geometry: an 8-torus has diameter 4 per axis, and a Feistel round moves information 2 hops, so χ_3 alone floods the cube in ≈ 3 rounds. Reach was never the problem. The tell was that the plateau was *flat and numerically exact*: a diffusion-reach limitation produces a slow ramp, not a hard ceiling at a suspiciously round number. **When a measurement lands on an exact value, compute what that value would have to mean.**

ρ fixes it: face-adjacent cells receive different rotation offsets, so bit k of a cell comes to depend on bit $k - \rho(j)$ of neighbour j, and the planes couple. This is precisely the job Keccak's ρ does, and it is why Keccak has a ρ at all. A regression test (`bitplanes_are_coupled`) now asserts that a single-bit difference reaches ≥ 2 bit-planes within 4 rounds.

5 Methodology

All code is Rust (stable, `-release`), with geometry lookup tables cached behind `OnceLock`; the table optimisation is verified behaviour-preserving by the fact that all 10 unit tests pass unchanged. Measured cost is $\approx 4.1\,\mu s$ per round per permutation. Four harnesses:

`avalanche` Strict avalanche ramp from a single flipped bit, averaged over 64 trials, reporting total/rate-shell/capacity-shell difference and the number of bit-planes reached.

`linearity_test` Superposition/affinity smoke test: for affine L, $L(a) \oplus L(b) \oplus L(0) \oplus L(a \oplus b) = 0$ identically.

`slide_test` Verifies the round constant breaks the period-3 self-similarity of the tap rotation.

`distinguish` Statistical distinguishers against the sponge's fixed-capacity slice: per-bit bias, single-bit differential propagation, and random small-weight linear approximations, reported as z-scores.

`attack` Cube attack with genuine key recovery: dependency-cone measurement, superpoly extraction, BLR linearity testing, GF(2) Gaussian elimination, verified against the real key.

`degree` Numeric-mapping degree bounds, soundness cross-check, and deterministic zero-sum search.

5.1 The attacked variant, and why the weakening is sound

`attack` and `degree` target a *deliberately weakened* variant:

$$\text{state} = \text{key}(16\text{B}) \,\|\, \text{nonce}(64\text{B}) \,\|\, \text{capacity}(0) \;\rightarrow\; \text{permute}_R \;\rightarrow\; \text{read rate},$$

whereas the shipped `keystream()` places a *full* permutation between the key absorb and the nonce absorb, so the attacker's nonce never meets the raw key. The variant is the standard Ascon/Keccak "init" model and is strictly easier to attack.

This direction of weakening is what makes the argument sound. An attack that dies at round R against the weaker variant implies the shipped construction survives *at least* R rounds:

a lower bound on the real thing's security, which is the useful direction for a defender. Claiming a *strong* result from a weakened variant would be illegitimate; claiming the conservative one is not.

5.2 An important non-experiment

We do not test "is any output bit of permute biased" on the full state. permute is a bijection on $\mathbb{F}_2^{4096}$, so uniform input yields exactly uniform output by definition; any measured bias is sampling noise. The meaningful question is the one an attacker faces — the pushforward through the sponge's *fixed-capacity slice*, which is not guaranteed uniform. All distinguishers target that slice.

6 Two harness bugs, and how they were caught

6.1 Non-monotonic degree

Our first `attack` reported estimated degrees of $0, 1, 0, 4$ for rounds 1..4. Degree cannot decrease with rounds under the naive bound, so this was impossible on its face. Cause: we drew random cubes from 2240 nonce bits and random output bits. At low rounds the permutation is *local*, so almost every cube sat entirely outside the output bit's dependency cone. The sums were zero because those nonce bits do not influence that output bit *at all* — we were measuring "did I hit the cone," not degree. Fix: measure the cones first, and draw cubes only from a target bit's own nonce-influence set.

6.2 Sampling a sparse target set

The corrected harness then reported "no target" at round 1. Cause: we sampled 24 of 2368 output bits, and at round 1 only $\approx 4\%$ of output bits have *any* key bit in their cone. Fix: full scan at rounds 1–2 (with the cheaper key-cone test as a filter before the nonce-cone test), sampling only at rounds ≥ 3 where cones have saturated. This is what turned rank 0 into actual recovered key bits.

7 Results

All results reproduced on Linux (Ubuntu, container) and Windows 11 (PowerShell 7.6.3) with different random seeds and secret keys, with agreement to within sampling noise.

7.1 Correctness

All 10 unit tests pass: per-round and full-permutation invertibility, ρ/π round-trips, π bijectivity ($|\{\pi(i)\}| = 512$), $\pi^{-1} \circ \pi = \mathrm{id}$, stream-cipher round-trip, shell partition sizes ($296 + 216 = 512$), checkerboard neighbour colouring, and bit-plane coupling.

7.2 Avalanche

Full avalanche is reached at round 6–7, so ROUNDS $= 12$ provides roughly a $2\times$ margin on this metric. Note that 5.94% at round 3 is now a mere waypoint; in the pre-ρ design it was the permanent ceiling. Critically, the **capacity shell tracks the rate shell within noise** (49.51% vs 49.93% at round 6), with no permanent lag: the concentric-shell rate/capacity split is doing real work rather than serving as decoration.

Table 1: Single-bit avalanche ramp, mean over 64 trials. The bit-plane column is the diagnostic for the bug of Section 4.2.

Rounds	Total diff	Rate shell	Capacity shell	Bit-planes hit (of 8)
0	0.02%	0.03%	0.02%	1.00
1	0.16%	0.17%	0.15%	4.70
2	1.08%	1.04%	1.13%	7.84
3	5.94%	5.94%	5.94%	8.00
4	23.14%	23.21%	23.05%	8.00
5	44.55%	44.61%	44.48%	8.00
6	49.75%	49.93%	49.51%	8.00
7	50.00%	49.95%	50.07%	8.00

7.3 Affinity smoke test

Residual weights (out of 4096 bits) over 200 trials per round count, at rounds $1, 2, 4, 8, 12$: minimum 1952–1975, mean 2029.6–2050.2, i.e. pinned at $\approx 50\%$ from round 1 onward. No trial ever produced an exact affine relation. This is a *necessary, not sufficient* check; it would fail loudly on a catastrophically linear design and says nothing about algebraic degree.

7.4 Slide symmetry

With round constants disabled, rounds 0 and 3 produce identical output on identical input (`true`); with constants enabled, they do not (`false`). Confirmed: the tap rotation alone yields a period-3 self-similar round function, and the round constant is exactly what breaks it.

7.5 Statistical distinguishers

The strongest signal was single-bit differential propagation (input bit 0 flipped, 20,000 trials, worst output bit reported):

Table 2: Differential propagation against the fixed-capacity slice. z is deviation from a 50% flip probability.

Rounds	Worst out-bit	z-score	Flip probability
1–3	—	± 141.42	$\approx 0\%$ or 100% (deterministic)
4	12	-141.42	$\approx 0\%$ (deterministic)
5	1798	-105.57	$\approx 12.7\%$
6	451	-18.38	$\approx 43.5\%$
7	183	-3.90	$\approx 48.6\%$
8	74	-3.54	$\approx 48.75\%$

Round 5 is trivially distinguishable. Round 6 is a real but small effect — $z = -18$ is not “maybe.” By round 7 it is at the noise floor. This independently corroborates the avalanche result (full mixing at 6–7) from a different measurement, which is the kind of internal consistency one wants before trusting either.

7.6 Cube attack: key recovery

Key recovery works reliably at round 1, intermittently at round 2, and never at round 3 or beyond. The cone table explains the mechanism precisely: a d-dimensional cube linearises degree $\approx d+1$, and degree grows $\approx 4^R$. At round 1 the cone is ≈ 3 bits and degree ≤ 4, so a $d = 3$ cube

Table 3: Dependency cones. Full scan of all 2368 output bits at rounds 1–2; sampling thereafter.

Rounds	Scan	Out-bits with key cone	Mean |key cone|	Mean |nonce cone|
1	full	253 of 2368	3.3	2.6
2	full	939 of 2368	6.5	9.3
3	sample	39 of 48	24.3	98.2
4	sample	48 of 48	80.5	332.6
5	sample	48 of 48	122.4	491.5

Table 4: Cube attack key recovery against a 128-bit key. Ranges are across four runs with different secret keys and seeds; cube selection is randomised, so R=2 succeeds in roughly half of runs. Every "bits pinned" figure was verified correct against the real key.

Rounds	Linear superpolys	Rank	Keyspace	Key bits pinned & verified
1	3–9	2–5	2^{123}–2^{126}	1–4 (all correct)
2	0–2	0–2	2^{126}–2^{128}	0–2 (all correct)
3	0	0	2^{128}	—
4	0	0	2^{128}	—
5	0	0	2^{128}	—

linearises trivially. By round 3 degree is ≈ 64, demanding 2^{64} work per superpoly. The attack does not fail because we gave up; it fails because 2^d outruns any budget the moment the cone opens.

7.7 Provable degree bounds and zero-sums

Numeric mapping was cross-validated for soundness before use: the bound claims 1336 output bits have degree 0 at round 1 (i.e. provably do not depend on *any* nonce bit). We sampled 120 such predictions and hammered them with random nonces: **0 violations**. Had even one moved, every number below would be worthless.

Table 5: Numeric-mapping degree bounds, all 512 nonce bits active. The bound is sound (an over-approximation) and beats the naive 4^R bound at round 4.

Rounds	Naive 4^R	Proven bound (max)	Median	Out-bits proven degree 0
1	4	4	0	1336 of 2368
2	16	16	3	176 of 2368
3	64	62	19	0
4	256	**151**	114	0
5	512	512 (sat.)	512	0

Proposition 2 (Zero-sum)**.** *If output bit o has provable degree $< d$ in a set of d cube variables, then summing o over all 2^d assignments of that cube is* 0 *— for every key, always.*

Proof. Every monomial of degree $< d$ omits at least one cube variable; summing over both values of an omitted variable cancels the monomial in pairs. □

This is a *deterministic* distinguisher: no statistics, no z-score. We verified sampled claims empirically and every predicted sum was indeed zero — theory and measurement agreeing.

Cube *shape* matters more than dimension. Compare "1 plane across 16 cells" (924 zero-sums) with "2 cells adjacent" (2360) — identical $d = 16$, a 2.5× difference in reach. The mechanism

Table 6: Cube *shape* sweep. Entries are output bits with a provable zero-sum. 0 means the bound was not tight enough to prove one, not that none exists.

Shape	d	Cost	R=4	R=5	R=6	R=7
1 cell, all 8 planes	8	2^8	2088	0	0	0
2 cells adjacent	16	2^{16}	2360	0	0	0
4 cells adjacent	32	2^{32}	2368	0	0	0
8 cells adjacent	64	2^{64}	2368	**136**	0	0
1 plane across 16 cells	16	2^{16}	924	0	0	0
1 plane across 32 cells	32	2^{32}	1580	0	0	0
2 planes × 12 cells	24	2^{24}	1335	0	0	0
Spread, stride-37	24	2^{24}	1230	0	0	0

follows directly from Section 4.2: χ_3 is bitwise, so bit k only meets bit k of its neighbours, and ρ is the sole coupling between planes. A cube stacked into few cells therefore grows degree *slower* than one spread across many, and slower growth means deeper zero-sums. The $d = 64$ cube reaches round 5 at 2^{64} work: unverifiable in practice, but $2^{64} \ll 2^{128}$, so it is a legitimate distinguisher — proven by propagation rather than measured.

8 Discussion

8.1 The ladder

Table 7: Complete cryptanalytic ladder against `cube3d-sponge` (12 rounds).

Result	Reach	Nature
Key recovery (cube attack)	2	Verified against real key
Provable zero-sum, practical ($\leq 2^{16}$)	4	Deterministic + empirically confirmed
Provable zero-sum, $\leq 2^{64}$ cube	5	Deterministic, proof-only
Statistical differential bias	6	$z = -18$; evidence, not proof
Design	**12**	$\approx 2\times$ margin over the best distinguisher

The four-round gap between key recovery (2) and distinguishing (6) is not an anomaly — it is the expected shape. Distinguishers always reach further than key recovery, which is why "I found a bias at round 5" never implies "the key falls at round 5."

8.2 Calibration against Ascon

The single most useful calibration point is Ascon's own third-party analysis table [24], which reports a **zero-sum distinguisher on the full 12/12-round permutation at** 2^{55}, an 8/12 integral at 2^{46}, and 5/12 truncated-differential and rectangle distinguishers. Ascon is nonetheless a NIST standard [25].

The lesson is important and cuts against alarmism: *distinguishers on the permutation do not break the mode.* A zero-sum on the bare permutation says nothing directly about the AEAD's confidentiality, because the mode never gives the attacker the freedom the distinguisher assumes. By that yardstick, our round-4/5 zero-sums on a 12-round permutation are unremarkable — Ascon has one at 12/12. What would be alarming is a *key recovery* near full rounds, and ours dies at 2.

8.3 What the geometry was worth

Three specific findings bear on the original question.

1. **Concentric shells as rate/capacity: validated.** The capacity shell tracks the rate shell within noise. The split does real work.
2. **Radiating outward as diffusion: refuted before it was built.** Toroidal wrapping, not radial structure, is what gives uniform neighbourhoods. Decoupling the diffusion geometry from the security boundary was necessary; the original proposal fused them.
3. **More shells for "more complexity": no effect, and mildly harmful.** Extra shells add state and cost, add no nonlinearity, and (without wrapping) add neighbourhood classes and hence anisotropy.

The deeper point is that *every* security-relevant property in the final design came from the correction list — nonlinearity (χ_3), invertibility (Feistel), symmetry-breaking (ι), plane coupling (ρ), transposition (π) — and none from the geometry. The cube is plumbing. Good plumbing, but plumbing.

9 Limitations

We state these bluntly because the failure mode of this genre is confident under-qualification.

1. **This is not a secure cipher and must not be used.** There is no authentication tag; it is a keystream generator. Use Ascon [25] or ChaCha20-Poly1305.
2. **Upper bounds cannot prove security.** Degree upper bounds prove distinguishers *exist*; they never prove none does. Every "0" in Table 6 means "this bound was not tight enough to prove a zero-sum," not "no zero-sum exists."
3. **Numeric mapping is the wrong tool; we used it for tractability.** Bit-based division property via MILP [30, 31] is strictly tighter [35], but our state is 4096 bits — 2.5× Keccak's 1600 and 12.8× Ascon's 320 — giving $\approx 4 \times 10^5$ binary variables and $\approx 3 \times 10^5$ constraints for a 5-round model, with no solver available. The gap is not academic: Hu et al. [34] show numeric mapping's Trivium bound reaches 79 at 793 rounds where the exact degree at 834 rounds is only 78. A proper BDP/monomial-prediction model on real hardware could plausibly push the proven reach past 5.
4. **Negative attack results are weak.** "No key bits at round 3" means this attack, at these cube dimensions, with this compute, found nothing. Real cube attacks use division-property degree bounds and clever cube selection [32], not random draws.
5. **No differential/linear trail bounds.** We have no proven bound on the maximum expected differential characteristic probability, which is table stakes for a real proposal.
6. **χ_3, ρ, π and RC were chosen for plausibility, not optimised.** The ρ offsets and the π shear were not searched for branch-number or diffusion optimality.
7. **No third-party analysis.** The design and all attacks share an author. This is the weakest point of all.

10 Conclusion and future work

We set out to answer whether 3D cellular automata on folded concentric-cube grids would make a cryptosystem "way more complex and hence more secure." The answer is that the second clause does not follow from the first, and that the geometric intuition — while pointing at genuinely the right shape, since Keccak is a 3D local nonlinear permutation — contributes nothing on its own. Security came entirely from a short, enumerable list of properties, every one of which had to be added deliberately and then *measured*.

The measurement is the contribution. Four harnesses, cross-validated on two operating systems, produce a coherent ladder: key recovery to round 2, deterministic zero-sums to round 4 (practical) or 5 (2^{64}), statistical bias to round 6, against a 12-round design. Two design bugs and two harness bugs were caught by numbers that were *impossible* rather than merely surprising — a flat plateau at exactly 6.25%, and a non-monotonic degree curve. We would emphasise that pattern above any specific result: build the falsifier first, and when it returns an exact number, work out what that number would have to mean.

Future work, in descending order of value: (i) a bit-based division property model using monomial prediction [34] on hardware capable of a 4096-bit state, to replace our sampled cones and loose bounds with provable ones; (ii) differential and linear trail bounds via MILP; (iii) a proper AEAD mode with a tag; (iv) optimisation of ρ/π for measured branch number. None of this would make the design trustworthy without independent cryptanalysis, which remains the only thing that ever does.

Availability

Source, all six harnesses, and the unit-test suite are available at `https://github.com/<user>/3d-ca-crypto`. Results reproduce with `cargo test -release` and `cargo run -release -bin <harness>`.

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